

# Demand System Asset Pricing (DSAP) and IV Identification

## 1. DSAP

Demand System Asset Pricing (DSAP) is a structural equilibrium framework that links **asset prices** to **who holds risk** (portfolios/holdings/flows) and **why** (beliefs, constraints, institutions, and fundamentals). The empirical targets are (i) **demand elasticities** and substitution patterns, and (ii) **demand shifters/shocks** (observed and latent). Once the demand system is estimated, **market clearing** turns it into a mapping from shocks to prices, enabling **equilibrium counterfactuals** (policy, regulation, segmentation, QE, etc.).

## 2. Why IV is essential: the endogeneity problem

A convenient workhorse specification for investor  $i$ 's demand for asset  $n$  is

$$Q_{int} = -\pi_{it} P_{nt} + \beta'_{it} X_{nt} + \varepsilon_{int}, \quad (1)$$

where  $P_{nt}$  is the price (or yield),  $X_{nt}$  are fundamentals/characteristics, and  $\varepsilon_{int}$  is *latent demand* (unobserved tastes/beliefs/constraints).

Market clearing implies prices depend on *aggregate* latent demand:

$$S_{nt} = \sum_{i=1}^I Q_{int} \Rightarrow P_{nt} = \frac{1}{\pi_t} (\beta'_t X_{nt} + \varepsilon_{nt} - S_{nt}), \quad (2)$$

so  $\text{Cov}(P_{nt}, \varepsilon_{int}) \neq 0$  in general. This is why DSAP typically needs an **instrument**: we must isolate price variation that is *not* driven by the focal investor's unobserved demand shock.

## 3. Identification logic

DSAP instruments are built around **residual supply**. Rewrite market clearing as

$$Q_{int} = S_{nt} - \sum_{j \neq i} Q_{jnt} \equiv RS_{int}.$$

To identify investor  $i$ 's demand elasticity, we need exogenous variation in  $RS_{int}$  (equivalently, in other investors' demand), which moves  $P_{nt}$  but is orthogonal to  $\varepsilon_{int}$ .

A valid instrument  $Z_{int}$  must satisfy:

- **Relevance:**  $Z_{int}$  shifts prices through residual supply, i.e.,  $\text{Cov}(Z_{int}, P_{nt}) \neq 0$ .
- **Exclusion:** conditional on controls,  $Z_{int}$  is independent of the focal latent-demand shock, i.e.,  $\text{Cov}(Z_{int}, \varepsilon_{int} | X_{nt}, \dots) = 0$ .

## 4. So is DSAP “basically finding IVs in micro finance data”?

**Not exactly.** IV is a *means* to identify demand elasticities when prices are endogenous, but DSAP is larger:

Microfoundation  $\rightarrow$  Demand estimation (often IV)  $\rightarrow$  Equilibrium mapping  $\rightarrow$  Counterfactuals.

That said, **credible DSAP work is very IV-intensive**: the best instruments are typically embedded in financial *institutions and rules* (benchmarks, mandates, regulation, purchase programs, custody, etc.). So yes, DSAP often rewards (i) strong IO-style IV skills and (ii) deep knowledge of market design.

## 5. Common DSAP IV

These instruments create plausibly exogenous residual-supply variation:

- **Investment mandates / restricted choice sets.** Assets exogenously included in many large investors’ “investment universes” have higher predicted demand, shifting prices independently of the focal investor’s shock.
- **Index reconstitutions and cutoff rules.** Mechanical membership rules generate sharp, plausibly exogenous demand shifts for close substitutes.
- **Policy purchases with predetermined rules (QE-style).** Purchase rules (e.g., proportional allocations) generate demand shifts that move prices while being plausibly excluded from asset-specific latent demand.
- **Predictable rebalancing shocks.** Pre-announced payouts or mechanical portfolio rules generate trading pressure that is not a contemporaneous response to prices.
- **Granular shocks.** Shocks to very large investors/sectors can be aggregated into instruments for the assets they disproportionately hold.

## 6. Separating fundamentals vs demand-driven price pressure

DSAP separates these conceptually via the equilibrium pricing map:

$$p_t = g(s_t, x_t, A_t, \theta_t, \varepsilon_t),$$

where  $x_t$  collects fundamentals/cash-flow and risk characteristics;  $A_t$  captures investor size/AUM (flows); and  $\varepsilon_t$  captures latent demand (beliefs/constraints/segmentation).

- **Fundamental-driven:** movements in  $x_t$  (cash flows, default risk, macro fundamentals).
- **Demand/flow-driven price pressure:** movements in  $A_t$  and  $\varepsilon_t$  that shift prices even when  $x_t$  is unchanged.

**Key point.** IV is what lets us cleanly estimate how much prices move when demand shifts. Once we have elasticities, we can run counterfactuals to quantify:

1. how much of a price move is consistent with fundamentals ( $x_t$ ), and
2. how much is attributable to flows / latent demand (price pressure).

## 7. How to position DSAP vs empirical IO vs corporate finance

**Empirical IO:** DSAP is “IO for asset markets” in spirit: estimate demand elasticities and run equilibrium counterfactuals. The extra structure is financial-market clearing and (often) high-dimensional substitution across assets/classes.

**Corporate finance:** typically identifies causal effects of firm decisions/shocks on real outcomes. DSAP instead targets equilibrium price formation given investor heterogeneity, constraints, and segmentation.

**Takeaway.** DSAP is where *institutional details* (market rules) and *IV discipline* jointly determine credibility.

## References

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